

Research

Assessing adversarial replay and deep learning-driven attacks on specific emitter identification-based security approaches

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Abstract

Specific Emitter Identification (SEI) has been put forward as an Internet of Things (IoT) Physical Layer Security (PLS) approach for its abilities to detect, characterize, and identify wireless emitters by exploiting distinct, inherent, and unintentional features in their transmitted signals. Since its introduction, a significant amount of work has been conducted; however, most assume the emitters are passive and their identifying signal features are immutable and challenging to mimic. This suggests the emitters are reluctant and incapable of developing and implementing effective SEI countermeasures; however, Deep Learning (DL) has been shown capable of learning emitter-specific features directly from their raw in-phase and quadrature signal samples, while Software-Defined Radios (SDRs) are capable of manipulating these signal samples. Based on these capabilities, it is fair to question the ease at which an emitter can effectively mimic the SEI features of another or manipulate its own to hinder or defeat SEI. This work considers SEI mimicry using three signal features mimicking countermeasures; “off-the-self” DL; two SDRs of different sizes, weights, power, and cost (SWaP-C); handcrafted and DL-based SEI processes, and assessment of an SEI mimicry countermeasure in a “coffee shop” deployment. Our results show “off-the-shelf” DL algorithms, and SDR enables SEI mimicry; however, adversary success is hindered by: (i) the use of decoy emitter preambles, (ii) the use of a denoising autoencoder and (iii) SDR SWaP-C constraints.

Article Highlights

- Assesses susceptibility of waveform-based radio identification security mechanisms to mimicry-based attackers.
- Use an adversary employing two software-defined radio form factors and readily available open-source machine learning.
- Shows the adversary sophistication needed to effectively impact waveform-based radio identification security mechanisms.

Keywords Specific emitter identification · RF-fingerprinting · IoT security · Internet of things · Adversaries · Penetration testing · Deep learning · Generative artificial intelligence

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1 Introduction

Specific Emitter Identification (SEI) was introduced almost 30 years ago to provide electronic warfare systems the capacity to detect, characterize, and identify radars by exploiting distinct, inherent, and unintentional features present in their transmitted signals [1]. Over that time, a high volume of SEI research has been conducted. More recently, SEI has seen a surge of interest due to Internet of Things (IoT) security concerns and the emergence of Deep Learning (DL). SEI is an advantageous IoT security approach because it passively exploits physical layer features unintentionally imparted to the signal—by the emitter’s Radio Frequency (RF) front-end circuitry—during its formation and transmission. Thus, SEI eliminates the need to modify or add IoT device functionality. Moreover, SEI’s passive nature makes it a viable security mechanism for resource-constrained IoT devices (e.g., limited memory, computation, or battery). Additionally, DL-based SEI has shown the capability to learn emitter-specific features directly from the raw In-phase and Quadrature (IQ) signal samples, which eliminates reliance on expert-driven feature engineering [2].

Most SEI research assumes the identified emitters are passive and their identifying features immutable and difficult to mimic, which suggests they are reluctant and incapable of developing and implementing effective SEI countermeasures [3, 4]. However, when considering DL’s ability to learn emitter-specific features directly from the raw IQ signal samples and coupling it with readily available “off-the-shelf” DL architectures and the flexibility of Software-Defined Radio (SDR), one must reconsider the degree to which these assumptions hold true [5, 6]. We assess an adversary’s ability to hinder or defeat an SEI-based IoT security approach by easily and effectively mimicking another emitter’s distinct, inherent, and unintentional signal features. The adversary does this by employing “off-the-shelf” DL to learn the signal features of another emitter and SDR to manipulate or mask its emitter-specific features. Prior works have investigated SEI mimicry [7–10]. However, our work differs from theirs and contributes to the community’s understanding of SEI mimicry in the following ways.

- Performs SEI mimicry using “off-the-shelf” DL algorithms while prior efforts use algorithms tailored specifically with SEI mimicry in mind. We aim to determine the ease at which SEI mimicry can be implemented.
- Conducts SEI mimicry using two SDRs that differ in Size, Weight, Power, and Cost (a.k.a., SWaP-C) to determine how SDR capabilities impact SEI mimicry effectiveness.
- Targets mimicry of non-SDR emitters. Specifically, the emitters being mimicked are eight TP-Link Archer T3U USB 802.11a Wireless-Fidelity (Wi-Fi) compliant emitters, better representing IoT deployed and user devices.
- Uses an adversary that conducts SEI mimicry using signal replay, an AutoEncoder (AE), or convolutional Generative Adversarial Network (GAN) to mimic the SEI features present within the signals of another emitter.
- Considers the presence of “decoy” emitter signals to aid the SEI process in discerning the adversary from the authorized emitter being mimicked. A decoy emitter is an SDR of the same manufacturer and model as that used by the adversary but has a different serial number.
- Assesses the effectiveness of GAN-based SEI mimicry within a realistic, in-situ “coffee shop” scenario unseen in previous works.
- Performs SEI and mimicry countermeasure assessments using handcrafted (a.k.a., feature-engineered) and DL-based processes to facilitate comparative assessment that is unseen in published literature.
- Develops the mimicry countermeasures directly from the collected signals’ raw In-phase and Quadrature (IQ) samples; thus, the employed mimicry countermeasure does not depend on nor is driven by a specific SEI implementation or process [e.g., handcrafted features with Support Vector Machines (SVM) versus raw IQ with a Convolutional Neural Network (CNN)].

Our results show “off-the-shelf” DL algorithms and SDR enables SEI mimicry; however, adversary success is hindered by (i) the use of decoy emitter signals, (ii) integration of a Denoising AE (DAE), and (iii) SDR SWaP-C constraints.

The article is organized as follows. Section 2 defines the threat model while Sect. 3 describes the signal of interest, RF-Distinct, Native Attributes (RF-DNA) Fingerprinting, and the machine learning architectures used. Section 4 describes the signal collection, detection, and preprocessing steps, DL configurations, and the “coffee shop” scenario. Section 5 presents the results and analysis while the conclusion is presented in Sect. 6.

2 Threat model

The threat model used herein extends the model presented in our prior work [11]. In that work, the adversary (a.k.a., Eve) attempts to gain unauthorized network access by assuming the digital identity (e.g., MAC address) of an authorized (a.k.a., Alice) device such that it is granted unauthorized network access by being incorrectly authenticated by the network monitor (a.k.a., Bob). It is assumed that Eve conducts the attack using simple software tools, is not an authorized device, and does not inherently have access to the network or its associated devices. Finally, it is assumed that network communication links and hardware are not originally compromised.

This work extends Eve's capabilities to include the use of "off-the-shelf" DL and Commercial-Off-The-Shelf (COTS) SDRs to develop and implement a mimicry countermeasure capable of hindering or defeating an SEI-based IoT security approach. Eve is implemented using two SDR platforms to determine SWaP-C impacts upon SEI mimicry success. The two SDR platforms are Ettus Research's Universal Software Radio Peripheral (USRP) B210 (~ \$4000 per unit) and Great Scott Gadget's HackRF One (~ \$470 per unit) [12, 13]. Eve's use of the full-duplex B210 is advantageous because it can receive its own transmitted signals while the half-duplex HackRF cannot. Thus, the HackRF requires Eve to employ a second one to receive the signals transmitted by the first, which increases Eve's complexity. Eve uses one of the following SEI mimicry countermeasures.

- **Replay SEI Mimicry:** Eve collects, saves, and re-transmits an authorized emitter's signals. Eve can modify the replayed signals by adjusting its transmit power and CFO behavior, but neither is assessed here.
- **AE-based SEI Mimicry:** Eve collects signals transmitted by Alice and trains an AE such that when Eve inputs one of its signals into the trained AE, it "recolors" the signal's features to match those present in Alice's signals. The AE has a hidden layer of size one hundred and a targeted Mean Squared Error (MSE) of 1×10^{-6} [14].
- **GAN-based SEI Mimicry:** Eve collects its own and Alice's signals. The GAN is trained by assigning Alice's collected preambles to Class #1 of the discriminator D and Eve's preambles to Class #2. During training, Class #2 is input to the generator G , and its weights are adjusted to learn the mapping needed to modify the SEI features—present in Eve's signals—to match those of Alice (a.k.a., Class #1) at the G 's output. The success of this mapping is determined by how often the D assigns the G 's output to Class #1. The D is a basic, three-layer CNN trained to discriminate Class #1's signals from those of Class #2 modified by the G . The G and D are iteratively updated until the D can no longer discern a Class #1 signal from those output by the G . Ideally, the G learns to remove Class #2's inherent SEI features and insert those of Class #1.

The AE and GAN are trained using unit energy signals to ensure the associated DL network learns the correct feature distributions without signal energy biasing the process. Only the signals' raw IQ representations are used to train the AE and GAN. The AE and GAN-based SEI mimicry countermeasures are implemented by passing the signal's IQ samples through the trained AE and G , and their outputs are stored for later transmission by Eve.

Eve employs each SEI mimicry countermeasure in conjunction with a falsified digital identity to increase its chance of being incorrectly authenticated by Bob. Bob verifies the identity of the to-be-authenticated device using the passed digital identity and the associated received signals' SEI features.

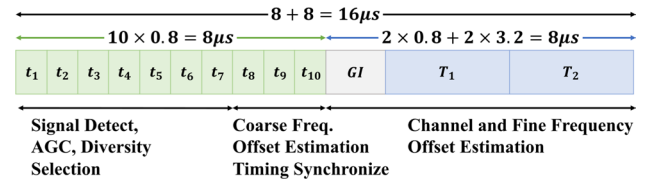
3 Background

3.1 Signal of interest

This work uses IEEE 802.11a Wi-Fi-compliant signals. IEEE 802.11a Wi-Fi is an Orthogonal Frequency Division Multiplexed (OFDM) digital encoding scheme. IEEE 802.11a Wi-Fi is used in this work for the following reasons.

- The first 16 μ s of every IEEE 802.11a Wi-Fi frame consists of a preamble. The preamble enables synchronization, carrier frequency and phase offset correction, and channel equalization at the receiver. An illustration of the IEEE 802.11a Wi-Fi preamble structure is shown in Fig. 1.

Fig. 1 The structure of the IEEE 802.11a Wi-Fi preamble that occupies the first 16 μ s of every transmitted signal [15]



- The IEEE 802.11a Wi-Fi preamble consists of a fixed and known sequence of symbols that makes it ideal for extracting steady-state signal features by an SEI process.
- The processes and techniques presented herein can be extended to other preamble-based standards such as ZigBee and Bluetooth.
- The number of publications demonstrating successful SEI exploitation of IEEE 802.11a Wi-Fi preamble extracted features [2, 14, 16–26]. These works have achieved serial number discrimination (i.e., same manufacturer and model number), which is the most challenging SEI case.
- The use of OFDM or a variation of OFDM (e.g., Orthogonal Frequency Division Multiple Access) in many digital communications schemes. These schemes include IEEE 802.11ac, IEEE 802.11ad, IEEE 802.11ax, and Long-Term Evolution (LTE).

3.2 RF-DNA fingerprinting

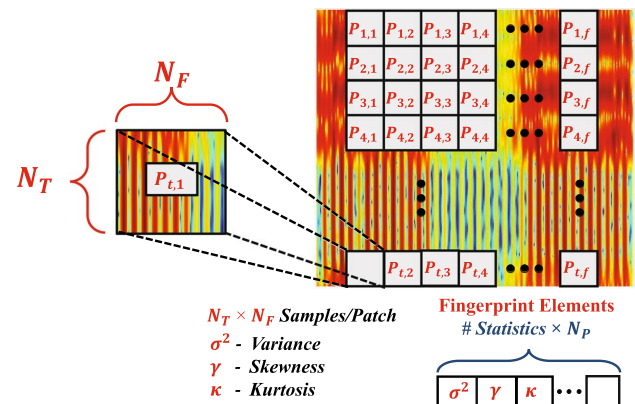
3.2.1 Handcrafted RF-DNA fingerprinting

Handcrafted RF-DNA fingerprints are extracted from the Wi-Fi preamble's two-dimensional (2D) Time-Frequency (TF) representation, which is generated using the Discrete Gabor Transform (DGT) [27],

$$G_{mk} = \sum_{n=1}^{MN_{\Delta}} s(n)W^*(n - mN_{\Delta})\exp^{-j2\pi kn/k_G}, \quad (1)$$

where G_{mk} are the complex-valued Gabor coefficients, $s(n) = s(n + lMN_{\Delta})$ is the periodic input signal, $W(n) = W(n - l mN_{\Delta})$ is the Gaussian window function, l is the period of the signal or window, the window is shifted N_{Δ} samples between calculations, $m = 1, 2, \dots, M$ for M total shifts, $k = 1, 2, \dots, K_G - 1$ for $K_G \geq N_{\Delta}$, and the modulus of (MN_{Δ}) with K_G equals zero [27]. The TF representation is generated by calculating the magnitude of the complex-valued Gabor coefficients, squaring the magnitude, and normalizing the result such that all of its values are in the range of zero to one. This surface is subdivided into N_p "patches", and the patch size is chosen to ensure that each patch contains at least fifteen elements. Each patch is converted into a one-dimensional (1D) vector and variance (σ^2), skewness (γ), and kurtosis (κ) calculated. This process is repeated for each patch over the entire TF surface, and the statistics are appended to those of the previous patch. Figure 2 shows the subdivision of the normalized, magnitude-squared Gabor-based surface into N_p patches, statistics calculated, and appended to form a handcrafted RF-DNA fingerprint. The three statistics are calculated over the entire TF surface and added to the end of the last patch's statistics. As in [28], the DGT is calculated with $M = 320$, $K_G = 320$, and $N_{\Delta} = 1$. Handcrafted RF-DNA fingerprints are generated using patches in which $N_T = 53$ and $N_F = 4$ that

Fig. 2 Gabor-based RF-DNA fingerprint generation is detailed in [21] and adopted for time-frequency-based SEI



results in each TF representation being divided into $N_p = 420$ patches; thus, resulting in $N_f = 1263$ features per hand-crafted RF-DNA fingerprint.

3.2.2 Deep learning-based RF-DNA fingerprinting

DL-based RF-DNA fingerprinting uses the preamble's time, frequency, or Gabor-based representation. The time and frequency representation is from our work in [14, 26]. For the time representation, the CNN learns RF-DNA fingerprints from the preamble's augmented IQ samples,

$$\mathbf{P}_{e,k} = \begin{bmatrix} i_{e,k}(1, 1) & i_{e,k}(1, 2) & \dots & i_{e,k}^i(1, N_s) \\ q_{e,k}(2, 1) & q_{e,k}(2, 2) & \dots & q_{e,k}(2, N_s) \\ \lambda_{e,k}(3, 1) & \lambda_{e,k}(3, 2) & \dots & \lambda_{e,k}(3, N_s) \\ \theta_{e,k}(4, 1) & \theta_{e,k}(4, 2) & \dots & \theta_{e,k}(4, N_s) \end{bmatrix}, \quad (2)$$

where k is the index assigned to the emitter's k th received preamble $\mathbf{r}_e$, N_s is the number of discrete-time samples comprising the emitter's k th preamble, i are the in-phase components, q are the quadrature components, λ and θ are the instantaneous magnitude and phase of $\mathbf{r}_e$. The frequency representation is generated using $\mathbf{r}_e$'s discrete Fourier transform's complex-valued coefficients and is expressed as,

$$\mathcal{P}_{e,k} = \begin{bmatrix} I_{e,k}(1, 1) & I_{e,k}(1, 2) & \dots & I_{e,k}(1, N_s) \\ Q_{e,k}(2, 1) & Q_{e,k}(2, 2) & \dots & Q_{e,k}(2, N_s) \\ \Lambda_{e,k}(3, 1) & \Lambda_{e,k}(3, 2) & \dots & \Lambda_{e,k}(3, N_s) \\ \Theta_{e,k}(4, 1) & \Theta_{e,k}(4, 2) & \dots & \Theta_{e,k}(4, N_s) \end{bmatrix}, \quad (3)$$

where I and Q are the real and imaginary components (a.k.a., the rectangular representation) of the complex-valued Fourier coefficients, respectively, Λ and Θ are the magnitude and phase responses (a.k.a., the polar representation) of the complex-valued Fourier coefficients, respectively.

The Gabor-based representation is the TF surface—in Sect. 3.2.1 before patch subdivision—converted into a Red, Green, and Blue (RGB) image that captures time, frequency, and color-based intensity in a three-dimensional tensor.

3.3 Machine learning architectures

3.3.1 Multiple discriminant analysis/maximum likelihood

Multiple Discriminant Analysis (MDA) reduces feature dimensionality while improving class separability. MDA extends Fisher's linear discriminant analysis from a two-class case to the C -class case, where C is the number of known emitters. MDA is a linear operation that projects the handcrafted RF-DNA fingerprints into a $(C - 1)$ -dimensional subspace without reducing class separability by maximizing inter-class distances while minimizing intra-class spread [29].

Maximum Likelihood (ML) classification assigns each MDA-projected RF-DNA fingerprint to one of the C possible classes using Bayesian Decision Theory. A classification decision is made by computing the likelihood value between the multivariate Gaussian distribution fit to each class and the *unknown* MDA-projected RF-DNA fingerprint. The *unknown* MDA-projected RF-DNA fingerprint is assigned to the class whose distribution results in the largest likelihood value. The classification error probability is minimized by assuming uniform costs and equal priors [29].

3.3.2 Convolutional neural network

A CNN is designed to classify image data; thus, its first layer extracts class-specific (a.k.a. emitter-specific) features by convolving 2D filters across an input image. This feature extraction layer is called a convolutional layer, and the learning process optimizes its filters. The convolutional layer's output is reduced by maximum pooling (a.k.a., MaxPooling), which saves the highest value enclosed by a window that is slid across the output. The resulting feature map is smaller but still preserves the most important data. MaxPooling is followed by a Rectified Linear Unit (ReLU) layer that zeros out any negative feature map values [30]. The use of a CNN is due to it being the most common DL architecture used in SEI as well as our familiarity with it [14, 26, 31–33].

3.3.3 Autoencoder

An AE is a deep generative model that attempts to reconstruct the input data at its output layer by learning a good representation and useful properties of the data. It can be described as consisting of an encoder and a decoder. The encoder generates a representation (a.k.a., a code) from the input data while the decoder regenerates the input data from the code [34]. In this work, an AE is trained by the adversary to learn the SEI feature distribution associated with Alice's signals. Once trained, Eve uses the AE to modify the SEI features present within its signals to match those of Alice.

3.3.4 Generative adversarial network

A GAN learns deep representations through semi-supervised or unsupervised learning with applications to image editing, style transfer, image synthesis, and classification [35, 36]. A GAN employs adversarial training to estimate a generative model by simultaneously training two networks: (i) a Generator G that captures the input data distribution and (ii) a Discriminator D that focuses on deciding if its input originated from the training data set or was created by the G . Thus, a GAN can be viewed as a minimax two-player game where G attempts to increase the probability that D makes the wrong decision. A unique solution to this minimax problem exists when (i) the G captures the training data distribution, and (ii) the probability that the D makes the correct decision is one-half everywhere [35]. As with the AE, Eve trains the GAN such that the G learns the SEI feature distribution of Alice's signals. Unlike the AE, the feedback relationship between the G and D networks allows the G to accept Eve's signals as inputs. After training, the G is disconnected from the D and integrated into Eve's architecture to modify its signal's SEI features to mimic Alice's.

4 Methodology

4.1 Signal collection, detection, and preprocessing

This section describes the SEI process' and Eve's signal collection and preprocessing procedures except those used in the "Coffee Shop" Scenario (see Sect. 4.3).

4.1.1 The SEI process

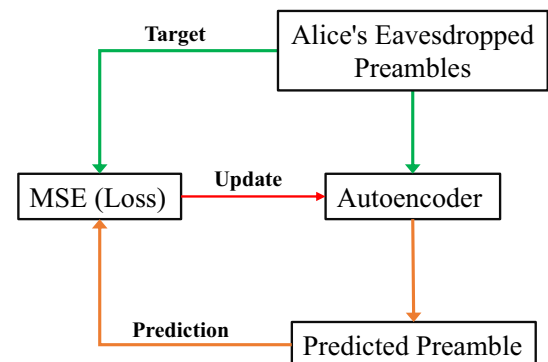
All signals are transmitted at 5.805 GHz and collected using a Tektronix Real-time Spectrum Analyzer (RSA) 5126B with an ultra wide-band antenna. The Device Under Test (DUT) (a.k.a., each of the TP-Links) transmits a 2 GB file to a Secure File Transfer Protocol (SFTP) server. The signals are sampled at 200 MHz, and the distance between the DUT and RSA antennas is set to achieve a Signal-to-Noise Ratio (SNR) of 30 dB or 9 dB, which corresponds to three or eight feet, respectively. Each collection is filtered using a fourth-order elliptical filter with a passband ripple of 0.5 dB, a stopband attenuation of 20 dB, and a cutoff frequency of 8.865 MHz. After filtering, an empirically selected amplitude threshold detects and removes individual Wi-Fi frames from each collection record. Each frame's preamble is detected by calculating the cross-correlation between the detected frame's magnitude and the magnitude of an ideal 802.11a preamble. The cross-correlation's maximum corresponds to the midpoint of the detected frame's preamble. Following detection and extraction, each preamble undergoes carrier frequency offset (CFO) correction according to [37], downsampling to 20 MHz, and energy normalization [26]. One thousand preambles represent each emitter.

4.1.2 The adversary

Eve continuously observes the 5.805 GHz channel and collects all signals using either a USRP B210 or HackRF SDR at a sampling rate of 40 MHz or 20 MHz, respectively. Individual preambles are detected and extracted from Eve's collection records using the approach described in Sect. 4.1.1. A total of one thousand preambles are collected and used to implement one of the three SEI mimicry countermeasures.

Table 1 Specifications of the AE used by Eve to create its corresponding SEI mimicking countermeasure

Layer #	Layer	Parameters	Activation
1	Dense	128	Sigmoid
2	Dense	1280	Sigmoid

Fig. 3 Flow diagram illustrating how Eve trains its AE-based mimicry countermeasure

4.2 Deep learning architecture configurations

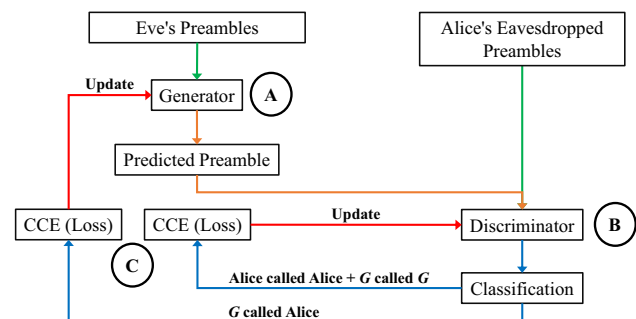
4.2.1 Mimicry networks

AE Mimicry: Eve uses the eavesdropped preambles from Alice as the input and target for a Multi-Layer Perceptron (MLP)-based AE whose parameters are given in Table 1. The AE utilizes the time-domain preamble's real, imaginary, magnitude, and phase components as the input and target output. Eve's AE consists of weights such that the hidden state is a vector of length 32, and the output is a vector of length 1280. Each layer is activated using the sigmoid function. The AE is trained until either (i) a target MSE of 10^{-8} has been reached or (ii) one hundred thousand epochs have been completed. The training process for the AE-based mimicry attack is illustrated in Fig. 3.

GAN Mimicry: Eve passes its own, self-monitored preambles through a convolutional generator G that is marked by Ⓐ in Fig. 4. The G , whose parameters are provided in Table 2(a), is comprised of (i) a two-dimensional Convolutional (2D-CONV) layer with sixty-four $[3 \times 3]$ filters, (ii) a 2D-CONV layer with one hundred and twenty-eight $[2 \times 4]$ filters, (iii) a MaxPooling layer with a $[2 \times 2]$ window, (iv) a 2D-CONV layer with sixty-four $[1 \times 3]$ filters, (v) a convolutional transpose layer with sixty-four $[3 \times 3]$ filters, (vi) a 2D upsampling layer of size $[1 \times 2]$, and (vii) a convolutional transpose layer with one $[1 \times 3]$ filters. The output of the G is a $[4 \times 320]$ tensor. The output of the G is passed to the convolutional discriminator D , whose parameters are detailed in Table 2(b), along with Alice's eavesdropped preambles. This step is marked in Fig. 4 by the Ⓑ. The D consists of (i) a 2D-CONV layer with sixty-four $[3 \times 3]$ filters, (ii) a 2D-CONV layer with one hundred and twenty-eight $[2 \times 3]$ filters, (iii) a MaxPooling layer with a $[2 \times 2]$ window, (iv) a 2D-CONV layer with thirty-two $[2 \times 3]$ filters, (v) a 2D-CONV layer with sixteen $[2 \times 5]$ filters, (vi) a flattening layer, (vii) a dense (a.k.a., a fully connected layer) with one-hundred and twenty-eight cells, and (viii) a dense layer with one cell. All G and D layers are activated using ReLU except for the two dense layers in the D . The D 's first dense layer has no activation, and its second dense layer is activated via the sigmoid function. The loss functions are calculated for each network (see Ⓒ in Fig. 4). The D aims to label Alice as one and the G output as zero. The G 's loss function is determined by calculating the Categorical Cross Entropy (CCE) when the D classifies the G output compared to a vector of ones (i.e., the D identifies Eve as Alice). The D 's loss function is determined by calculating the CCE when the D predicts Alice's preambles compared to a vector of ones (i.e., the D identifies Alice's preambles as originating from Alice), calculating the CCE when the D predicts the output of the G compared to a vector of zeros (i.e., the D identifies the G 's output as coming from the G), and adding the two values. The G and D are trained on the real, imaginary, magnitude, and phase representation of the time-domain preambles. The GAN is trained until ten epochs have been reached with a 250-sized minibatch.

Table 2 Specifications for the GAN architecture used by Eve to mimic the SEI features present in Alice's preambles

(a) Network parameters for the generator, <i>G</i>			
Layer #	Layer	Parameters	Activation
1	Input - Eve's RF-DNA Fingerprint	$[4 \times 320]$	
2	2D-CONV	$64-[3 \times 3]$	ReLU
3	2D-CONV	$128-[2 \times 4]$	ReLU
4	MaxPooling	$[2 \times 2]$	
5	2D-CONV	$64-[1 \times 3]$	ReLU
6	Transpose	$64-[1 \times 3]$	ReLU
7	2D Upsample	$[1 \times 2]$	ReLU
8	Transpose	$[1 \times 3]$	Sigmoid
9	Output - Generated RF-DNA Fingerprint	$[4 \times 320]$	Sigmoid
(b) Network parameters for the discriminator, <i>D</i>			
Layer #	Layer	Parameters	Activation
1	Input - Eve's Generated or Alice's Collected RF-DNA Fingerprint	$[4 \times 320]$	
2	2D-CONV	$64-[3 \times 3]$	ReLU
3	2D-CONV	$128-[2 \times 4]$	ReLU
4	MaxPooling	$[2 \times 2]$	
5	2D-CONV	$32-[2 \times 3]$	ReLU
6	2D-CONV	$16-[2 \times 5]$	ReLU
7	Flatten		
8	Dense	128	
9	Dense	1	Sigmoid

Fig. 4 Flow diagram illustrating how Eve trains its GAN-based mimicry counter-measure

4.2.2 Configurations of the DL-based SEI processes

Time- or frequency representation-based SEI: The specifications of the CNNs that perform SEI using either the time, Eq. (2), or frequency, Eq. (3), representations are provided in Table 3a. The corresponding CNN consists of three consecutive 2D-CONV layers for both representations and a $[4 \times 320]$ input tensor. All 2D-CONV layers use a $[4 \times 4]$ filter, and the number of filters per convolutional layer is forty, eighty, and one-hundred sixty, respectively. The first and second 2D-CONV layers are normalized using a ReLU layer, followed by MaxPooling using a patch size of four and a stride of four. All networks are trained using one hundred epochs with an 8000-sized minibatch.

Gabor image-based SEI: For DL-based SEI using Gabor images, the CNN is comprised of three layers and accepts a $[320 \times 320 \times 3]$ input tensor, Table 3(b). The input's third dimension captures the RGB colors of the Gabor-derived image. All 2D-CONV layers have a $[32 \times 32]$ filter size. The first 2D-CONV layer consists of eight filters, while the second and third layers contain sixteen and thirty-two filters, respectively. Each layer uses MaxPooling normalization with a $[32 \times 32]$ patch size, a stride of thirty-two, and ReLU activation. Each network is given thirty training epochs and an 8000-sized minibatch. Regardless of input representation (a.k.a., time, frequency, or Gabor images), the classification layer of the corresponding CNN is a dense layer comprised of eight cells when a decoy emitter *is not* used during training and nine

Table 3 Specifications for the CNN used to perform SEI

(a) Specifications for the CNN used to perform time and frequency representation-based SEI

Layer #	Layer	Parameters	Activation
1	Input - Time or Frequency Preamble Representation	$[4 \times 320]$	
2	2D-CONV	40— $[4 \times 4]$	ReLU
3	MaxPooling	$[4 \times 4]$	ReLU
4	2D-CONV	80— $[4 \times 4]$	
5	MaxPooling	$[1 \times 1]$	ReLU
6	2D-CONV	160— $[4 \times 4]$	ReLU
8	Dense	8, 9	Softmax

(b) Specifications for the CNN used to perform Gabor image-based SEI

Layer #	Layer	Parameters	Activation
1	Input - Gabor Image	$[320 \times 320]$	
2	2D-CONV	8— $[32 \times 32]$	ReLU
3	MaxPooling	$[32 \times 32]$	
4	2D-CONV	16— $[32 \times 32]$	ReLU
5	MaxPooling	$[8 \times 8]$	
6	2D-CONV	32— $[32 \times 32]$	ReLU
8	Dense	8,9	Softmax

when a decoy emitter *is* used to train the CNN. The eight and nine-cell dense layers are both activated using softmax and use CCE as their loss function. All CNN parameters are optimized using the ADAPtive Momentum (ADAM) algorithm with a learning rate of 0.01, a gradient threshold of 1, and a Ridge (a.k.a., L2) regularization factor of 10^{-4} . The CNNs are trained using MATLAB®'s DL Toolbox across four compute nodes, each with two AMD Epyc 7662 64C/128T CPUs and an Nvidia A100 80GB PCI-E card running on CUDA 12.0. Batch normalization is implemented after each convolutional layer to unify each Graphical Processing Unit's (GPU's) training data scale.

4.3 The “coffee shop” scenario

For this scenario, the “coffee shop” is our laboratory with dimensions of $6.7 \times 9.9 \times 2.8$ m ($21.98 \times 32.48 \times 9.19$ ft.), which represents a coffee shop with a small seating area. All emitters and the Access Point (AP) are static in this scenario, which is realistic for coffee shoppers who may be using their devices (e.g., laptop, tablet, phone) while enjoying a cup of coffee. This work does not consider a dynamic scenario (i.e., mobile Alice, Bob, or Eve) because environmental and channel changes make it more difficult for Bob and Eve to extract consistent/stable SEI exploited features from the collected signals due to multipath and Doppler effects, which may increase the possibility of Bob incorrectly identifying Eve as Alice and increase Eve's success in inhibiting or defeating the SEI-based security mechanism. Additionally, prior SEI works successfully extract consistent and stable features from signals collected under dynamic scenarios [38–40].

Figure 5 shows the setup used for this scenario, which consists of an IEEE 802.11a Wi-Fi network comprised of three authorized emitters—designated User A, User B, and User C—and an AP. Each user is a TP-Link AC1300 USB Wi-Fi adapter and a computer running Ubuntu Linux 16.0. The AP consists of an ASUS RT-AX56U IEEE 802.11a Wi-Fi router and a Nuand BladeRF SDR powered by a Raspberry Pi 4 model B. The AP is roughly equidistant, see Table 4, from each of the three authorized emitters. The Nuand BladeRF SDR collects the signals transmitted by each emitter and performs SEI using a CNN and the time representation of Eq. (2) to authenticate a given user's identity. An SFTP server provides a file transfer service for users in the coffee shop network. The SFTP server is located at User C. User A and User B are configured to transmit binary files to User C, and User C transmits binary files to User A. The AP forwards Wi-Fi frames from User A and User B to the SFTP server, while the BladeRF SDR collects the same frames to train the CNN for SEI.

Eve uses a B210 or two HackRFs controlled by an NVIDIA Jetson Nano Developer Kit. Regardless of the SDR, Eve (i) repeatedly transmits the same frame from Location D or E, (ii) receives Wi-Fi frames transmitted by itself and the three users, (iii) processes all Wi-Fi frames as described in Sect. 4.1, (iv) trains the GAN using backpropagation on an NVIDIA

Fig. 5 The coffee shop scenario setup in our laboratory for in-situ testing of Eve’s GAN-based SEI mimicry countermeasure. The laboratory is a $6.7 \times 9.9 \times 2.8$ m ($21.98 \times 32.48 \times 9.19$ ft.) room; thus, representing a small coffee shop. The red arrows indicate the user emitter whose SEI features are being mimicked by Eve from its indicated location

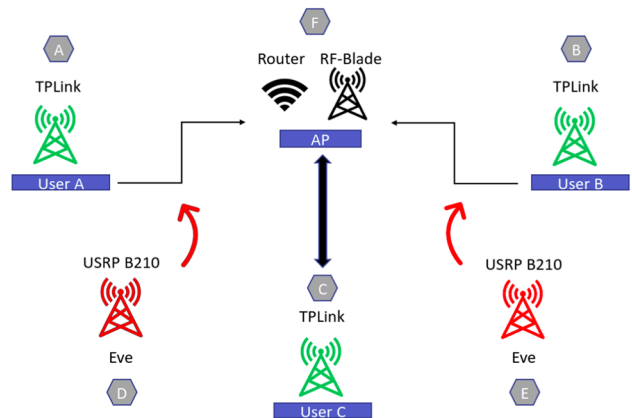


Table 4 The distances, in meters (m), between emitters and the access point (AP) used in the “coffee shop” scenario

	AP	User A	User B	User C
AP	–	1.55 m	1.55 m	1.4 m
Eve at location D	1.9 m	1.4 m	1.3 m	–
Eve at location E	1.9 m	–	1.4 m	1.3 m

Table 5 Identity authentication outcomes

Actual ID	System declaration	
	Authorized	Adversary
Authorized	True authentication (TAR)	False reject (FRR)
Adversary	False authentication (FAR)	True reject (TRR)

Tesla K40m GPU and quantizes the model using 12 bit resolution, and (v) uses the trained G to attack the SEI process at Location F. Eve mimics User A’s and User C’s SEI features from Location D while User B’s SEI features are mimicked when Eve is at Location E. Eve’s distances from each user and the AP are provided in Table 4. Eve has a Line-Of-Sight (LOS) with all users’ emitters and the AP. Lastly, the AP is far enough away from Eve and the three users to create unique channels between itself and each of them because a distance greater than a wavelength, which is 51.6 mm in our case, results in a unique channel between any transmitter-receiver pair [41]. During its attack, Eve removes the last two rows of the G ’s output tensor because they are the magnitude and phase of the IQ samples and are no longer needed. Ten thousand preambles are collected for User A, User B, User C, and Eve when its SEI countermeasure is turned off. Eight thousand preambles are randomly selected for training and validation, while the rest are used for testing. The collected preambles have an estimated SNR ranging from 10 to 12 dB. It is worth noting that none of the collected signals undergo channel correction; thus, any location-specific channel characteristics remain and are present in the signals processed by Eve and Bob.

5 Results

Eve’s SEI mimicry countermeasures are evaluated using the outcomes in Table 5. Of particular interest are the False Authentication Rate (FAR) and True Reject Rate (TRR), which indicate the rate at which Bob incorrectly authenticates Eve as an authorized emitter and correctly identifies Eve as an unauthorized emitter. FAR is the number of times Eve is incorrectly authenticated as one of the authorized emitters divided by the total number of transmissions made by Eve. The TRR is the number of times Eve is correctly identified as an unauthorized emitter divided by the number of transmissions made by Eve. Since SEI processes perform classification, the TRR will be 0% whenever the RF-DNA

Table 6 Eve's mimicry countermeasures versus preamble extracted SEI: FAR and TRR—presented in each cell as “FAR/TRR” using percentages—when each SEI process is and is not trained using a decoy emitter's RF-DNA fingerprints at SNR values of 30 dB and 9 dB

(a) SEI without the decoy present at SNR = 30 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	12.20%/00.00%	10.49%/00.00%	01.94%/00.00%	08.94%/00.00%	12.50%/00.00%
	AE	12.20%/00.00%	07.60%/00.00%	00.18%/00.00%	05.21%/00.00%	13.45%/00.00%
	GAN	12.20%/00.00%	13.24%/00.00%	12.58%/00.00%	21.09%/00.00%	12.50%/00.00%
B210	RE	12.50%/00.00%	12.50%/00.00%	12.71%/00.00%	13.39%/00.00%	12.50%/00.00%
	AE	12.50%/00.00%	13.98%/00.00%	20.96%/00.00%	20.19%/00.00%	01.03%/00.00%
	GAN	12.50%/00.00%	07.33%/00.00%	05.79%/00.00%	10.41%/00.00%	16.10%/00.00%

(b) SEI with the decoy present at SNR = 30 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	12.58%/00.00%	12.50%/00.00%	12.39%/05.65%	00.99%/78.95%	04.63%/76.55%
	AE	12.58%/00.00%	12.50%/00.00%	04.05%/42.03%	00.00%/100.00%	00.00%/98.36%
	GAN	12.58%/00.00%	12.50%/00.00%	12.41%/36.66%	00.00%/100.00%	00.00%/99.88%
B210	RE	11.75%/01.71%	00.00%/100.00%	16.98%/05.71%	10.16%/04.39%	12.13%/00.05%
	AE	11.75%/01.71%	00.00%/100.00%	12.66%/41.84%	00.00%/98.95%	00.00%/75.08%
	GAN	11.75%/01.71%	00.00%/100.00%	08.63%/31.80%	00.01%/99.96%	00.95%/89.46%

(c) SEI without the decoy present at SNR = 9 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	09.29%/00.00%	12.50%/00.00%	07.91%/00.00%	14.28%/00.00%	08.21%/00.00%
	AE	09.29%/00.00%	12.50%/00.00%	00.53%/00.00%	08.78%/00.00%	02.99%/00.00%
	GAN	09.29%/00.00%	12.50%/00.00%	10.79%/00.00%	09.28%/00.00%	07.49%/00.00%
B210	RE	18.93%/00.00%	15.00%/00.00%	15.25%/00.00%	24.23%/00.00%	21.44%/00.00%
	AE	18.93%/00.00%	08.19%/00.00%	02.59%/00.00%	14.36%/00.00%	10.08%/00.00%
	GAN	18.93%/00.00%	12.49%/00.00%	15.06%/00.00%	13.41%/00.00%	20.23%/00.00%

(d) SEI with the decoy present at SNR = 9 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	07.28%/38.81%	12.50%/00.00%	00.05%/74.79%	07.11%/41.06%	01.55%/78.80%
	AE	07.28%/38.81%	12.50%/00.00%	00.03%/96.15%	00.18%/97.80%	00.00%/99.59%
	GAN	07.28%/38.81%	12.50%/00.00%	00.01%/99.55%	00.56%/81.80%	00.01%/99.00%
B210	RE	17.71%/04.20%	12.50%/00.00%	08.09%/14.95%	23.18%/02.73%	21.44%/02.43%
	AE	17.71%/04.20%	08.38%/00.00%	00.79%/56.79%	01.04%/69.34%	01.49%/80.03%
	GAN	17.71%/04.20%	17.78%/00.00%	09.25%/45.73%	06.03%/67.93%	03.63%/92.65%

SEI processes that result in a FAR above 12.5% (a.k.a., a guess) and a TRR above 90% are marked using italicized and bold font, respectively. “RE”, “AE”, and “GAN” indicate Eve's replay, AE, and GAN-based mimicry countermeasures

fingerprints of a decoy emitter *are not* part of the training set (i.e., Eve is unknown to the SEI process before its attack) because the decoy represents Eve and serves as the unauthorized emitter. This is a byproduct of the classification process, which assigns every input RF-DNA fingerprint to one of the known classes no matter how poorly the RF-DNA fingerprint matches the class to which it is assigned. This classification drawback is addressed by either including

Table 7 Eve's mimicry countermeasures versus residual preamble extracted SEI: FAR and TRR—presented in each cell as “FAR/TRR” using percentages—when each SEI process is and is not trained using a decoy emitter's RF-DNA fingerprints at SNR values of 30 dB and 9 dB

(a) SEI without the decoy present at SNR = 30 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	13.06%/00.00%	16.74%/00.00%	12.31%/00.00%	14.35%/00.00%	04.79%/00.00%
	AE	09.30%/00.00%	12.94%/00.00%	14.39%/00.00%	18.39%/00.00%	09.66%/00.00%
	GAN	03.14%/00.00%	11.35%/00.00%	13.44%/00.00%	12.64%/00.00%	11.69%/00.00%
B210	RE	07.93%/00.00%	10.49%/00.00%	32.65%/00.00%	15.50%/00.00%	02.49%/00.00%
	AE	07.70%/00.00%	04.05%/00.00%	02.21%/00.00%	11.01%/00.00%	16.76%/00.00%
	GAN	28.60%/00.00%	19.73%/00.00%	11.93%/00.00%	13.65%/00.00%	10.25%/00.00%

(b) SEI with the decoy present at SNR = 30 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	04.58%/19.40%	06.90%/67.60%	00.08%/96.15%	00.78%/94.08%	05.24%/55.58%
	AE	01.43%/67.14%	00.68%/76.11%	00.24%/77.91%	01.46%/93.14%	00.19%/86.93%
	GAN	02.35%/61.45%	00.00%/98.06%	00.83%/91.64%	00.40%/95.64%	00.60%/92.85%
B210	RE	07.24%/03.65%	00.21%/58.54%	21.48%/11.69%	10.94%/15.41%	01.81%/00.06%
	AE	03.05%/49.24%	00.01%/83.96%	01.86%/78.16%	01.89%/85.69%	00.06%/79.99%
	GAN	23.33%/03.31%	05.64%/56.21%	00.86%/87.54%	02.01%/90.51%	00.96%/81.41%

(c) SEI without the decoy present at SNR = 9 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	09.53%/37.54%	12.28%/02.64%	04.35%/39.19%	09.31%/37.69%	02.24%/66.56%
	AE	09.49%/71.91%	13.30%/05.95%	02.81%/60.83%	05.49%/68.89%	00.09%/97.84%
	GAN	00.99%/75.01%	11.63%/02.94%	02.85%/65.78%	03.36%/69.40%	00.23%/95.19%
B210	RE	14.33%/11.45%	10.48%/04.04%	11.11%/16.81%	14.03%/03.28%	20.08%/02.35%
	AE	10.46%/41.93%	06.50%/09.30%	02.40%/62.41%	06.78%/46.34%	03.55%/75.29%
	GAN	13.16%/04.08%	19.63%/04.34%	02.65%/82.15%	07.65%/39.56%	02.99%/92.01%

(d) SEI with the decoy present at SNR = 9 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	09.53%/37.54%	12.28%/02.64%	04.35%/39.19%	09.31%/37.69%	02.24%/66.56%
	AE	09.49%/71.91%	13.30%/05.95%	02.81%/60.83%	05.49%/68.89%	00.09%/97.84%
	GAN	00.99%/75.01%	11.63%/02.94%	02.85%/65.78%	03.36%/69.40%	00.23%/95.19%
B210	RE	14.33%/11.45%	10.48%/04.04%	11.11%/16.81%	14.03%/03.28%	20.08%/02.35%
	AE	10.46%/41.93%	06.50%/09.30%	02.40%/62.41%	06.78%/46.34%	03.55%/75.29%
	GAN	13.16%/04.08%	19.63%/04.34%	02.65%/82.15%	07.65%/39.56%	02.99%/92.01%

SEI processes that result in a FAR above 12.5% (a.k.a., a guess) and a TRR above 90% are marked using italicized and bold font, respectively. “RE”, “AE”, and “GAN” indicate Eve's replay, AE, and GAN-based mimicry countermeasures

decoy emitter RF-DNA fingerprints (see Tables 6 and 7) or Eve's unmodified signals (see Sect. 5.3 the "Coffee Shop Scenario") within the SEI processes' training sets.

5.1 Results: Eve versus preamble extracted SEI

Table 6 shows FAR and TRR—presented in each cell as "FAR/TRR" using percentages—when each SEI process is and is not trained using a decoy emitter's RF-DNA fingerprints at an SNR of 30 dB or 9 dB. MDA/ML-based SEI uses either Gabor-based RF-DNA fingerprints (see Sect. 3.2.1) or the frequency representation of Eq. (3). The CNN-based SEI learns RF-DNA fingerprints from the time, frequency, or Gabor-based images described in Sect. 3.2.2.

At an SNR = 30 dB, Table 6(a) shows that a HackRF-based Eve employing replay and AE-based mimicry is classified at a FAR = 12.5% by the CNN-based SEI using all three preamble representations. When the HackRF-based Eve uses GAN-based mimicry, the FAR increases to 13.24% and 12.58% for the frequency and Gabor image representations, respectively. The FAR = 13.45% when an MDA/ML-based SEI process—using the frequency representation—classifies the HackRF-based Eve employing AE-based mimicry. When Eve uses the HackRF and GAN-based mimicry, the MDA/ML-based SEI process classifies Eve at a FAR = 21.09% when using handcrafted Gabor-based RF-DNA fingerprints. A B210-based Eve employing replay mimicry results in a FAR = 12.71% when countering a CNN-based SEI process that extracts emitter-specific features from Gabor images. When the B210-based Eve employs the AE-based mimicry countermeasure, a FAR of 13.98% and 20.96% is achieved when the CNN-based SEI process extracts emitter-specific features from the frequency and Gabor image representations, respectively. This same Eve achieves a FAR of 20.19% when countering the MDA/ML-based SEI process that classifies emitters using handcrafted Gabor-based RF-DNA fingerprints. When Eve uses a B210 in conjunction with a GAN-based mimicry countermeasure, a FAR of 16.10% is achieved versus the MDA/ML-based SEI process that classifies emitters using the frequency representation of Eq. (3).

Table 6(b) shows FAR and TRR for preambles collected at 30 dB and SEI processes trained *with* decoy emitter preambles represented within the training sets. A HackRF-based Eve using replay, AE, or GAN-based mimicry countermeasures achieves a FAR = 12.58% against the CNN-based SEI process that extracts RF-DNA fingerprints from the time representation, Eq. (2). A HackRF-based Eve using AE-based mimicry is defeated at a TRR of 100% and 98.36% when MDA/ML-based SEI uses handcrafted Gabor and frequency-based RF-DNA fingerprints to classify the emitters. When Eve switches from an AE-based to GAN-based mimicry countermeasure, the MDA/ML-based SEI process achieves a TRR of 100% and 99.88% when using handcrafted Gabor and frequency-based RF-DNA fingerprints to classify emitters. When Eve uses a B210 and any of the three mimicry countermeasures, the CNN-based SEI process rejects all of them at a TRR = 100% when learning RF-DNA fingerprints from the frequency representation. However, the TRR is poor when the CNN-based SEI process extracts RF-DNA fingerprints from the Gabor images with FARs of 16.98% and 12.66% resulting when a B210-based Eve employs replay and AE-based mimicry countermeasures, respectively. A B210-based Eve using either an AE or GAN-based mimicry countermeasure is successfully rejected at TRRs of 98.95% and 99.96% when the MDA/ML-based SEI process classifies the emitters using their handcrafted Gabor-based RF-DNA fingerprints.

Table 6(c) shows FAR and TRR for preambles collected at 9 dB and SEI processes trained *without* decoy emitter preambles represented in the training set. A HackRF-based Eve using the replay mimicry countermeasure achieves a FAR = 14.28% against the MDA/ML-based SEI process that performs emitter classification using handcrafted Gabor-based RF-DNA fingerprints. A HackRF-based Eve does not achieve a FAR 12.5% for all other mimicry countermeasures and SEI processes. A B210-based Eve employing the replay-based mimicry countermeasure achieves FARs of 18.93%, 15.00%, and 15.25% against the CNN-based SEI processes that extract RF-DNA fingerprints from the time, frequency, and Gabor image representations, respectively. This same Eve configuration achieves a FAR of 24.23% and 21.44% against the MDA/ML-based SEI processes that classify emitters using the handcrafted Gabor RF-DNA fingerprints and frequency representation, respectively. A FAR = 18.93% is achieved by a B210-based Eve that employs an AE-based mimicry countermeasure against the CNN-based SEI process that extracts RF-DNA fingerprints from the time representation, Eq. (2). When employing an AE-based mimicry countermeasure against an MDA/ML-based SEI process using handcrafted Gabor-based RF-DNA fingerprints, a B210-based Eve achieves a FAR = 14.36%. The B210-based Eve achieves a FAR of 18.93% and 15.06% when using a GAN-based mimicry countermeasure against the CNN-based SEI processes that extract RF-DNA fingerprints from the time and Gabor image representations. This same Eve also achieves FAR values of 13.41% and 20.23% against the MDA/ML-based SEI processes that use handcrafted Gabor and frequency-based RF-DNA fingerprints to classify the emitters.

Table 6(d) shows FAR and TRR for preambles collected at 9 dB and SEI processes that are trained *with* decoy emitters represented within the training set. A HackRF-based Eve using an AE or GAN-based mimicry countermeasure is successfully rejected at a TRR of 96.15% or 99.55% by the CNN-based SEI process that learns RF-DNA fingerprints from the Gabor images. The MDA/ML SEI process also rejects a HackRF-based Eve employing an AE-based mimicry countermeasure at a TRR of 97.8% and 99.59% when using handcrafted Gabor and frequency-based RF-DNA fingerprints to classify the emitters. When using frequency-based RF-DNA fingerprints, the MDA/ML SEI process rejects a HackRF-based Eve employing the GAN-based mimicry countermeasure at a TRR = 99%. All three mimicry countermeasures achieve a FAR of 17.71% when Eve employs a B210 against the CNN-based SEI process that extracts RF-DNA fingerprints from the time representation. A FAR of 17.78% results when a B210-based Eve uses the GAN-based mimicry countermeasure against the CNN-based SEI process process that extracts RF-DNA fingerprints from the preambles' frequency representation. A FAR of 23.18% and 21.44% is achieved by the B210-based Eve employing the replay-based mimicry countermeasure against the MDA/ML-based SEI process that classifies emitters using their handcrafted Gabor-based RF-DNA fingerprints. The MDA/ML-based SEI process that classifies emitters using the preambles' frequency representation rejects a B210-based Eve employing the GAN-based mimicry countermeasure at a TRR = 92.65%.

5.2 Results: Eve versus residual preamble extracted SEI

The structure and entries of Table 7 are consistent with Table 6 in which each cell contains the FAR and TRR percentages presented as "FAR/TRR". Results are shown for the cases when each SEI process is and is not trained using a decoy emitter's RF-DNA fingerprints at an SNR of 30 dB or 9 dB. The MDA/ML and CNN-based SEI results are generated using the same representations used to generate the results in Table 6 of Sect. 5.1. Unlike the results in Sect. 5.1 that are generated using the emitters' preambles, the results in this section are generated using the emitters' residual preambles or their representations (time, frequency, Gabor, etc.). The residual preamble is generated by subtracting an ideal preamble from each received preamble as detailed in [33]. The use of the residual preamble is motivated by the fact that the IEEE 802.11a Wi-Fi preamble's intentional, information-carrying structure provides no emitter-specific information to the SEI processes and could even hinder the SEI processes' ability to discriminate between emitters by acting as a confuser.

Table 7(a) shows FAR and TRR for residual preambles collected at 30 dB and SEI processes trained *without* decoy emitter residual preambles represented within the training set. When using the replay mimicry countermeasure, a HackRF-based Eve achieves a FAR of 13.06% and 16.74% against the CNN-based SEI process that extracts RF-DNA fingerprints from the time or frequency representation. When employing an AE or GAN-based mimicry countermeasure, a HackRF-based Eve achieves FARs of 14.39% and 13.44% against the CNN-based SEI process that extracts RF-DNA fingerprints from Gabor images. The HackRF-based Eve also achieves FARs of 14.35%, 18.39%, and 12.64% when employing replay, AE, and GAN-based mimicry countermeasures against the MDA/ML-based SEI process that classifies emitters using the handcrafted Gabor-based RF-DNA fingerprints. A B210-based Eve achieves a FAR of 32.65% when using the replay mimicry countermeasure against the CNN-based SEI process that extracts RF-DNA fingerprints from the Gabor images. When using the GAN-based mimicry countermeasure, the B210-based Eve achieves FARs of 28.60% and 19.73% against the CNN-based SEI process that extracts RF-DNA fingerprints from the residual preambles' time or frequency representation. Replay and GAN-based mimicry countermeasures, employed by a B210-based Eve, achieve FARs of 15.5% and 13.65% against the MDA/ML-SEI process that classifies emitters using handcrafted Gabor-based RF-DNA fingerprints. The B210-based Eve achieves a FAR = 16.76% when employing the AE-based mimicry countermeasure against the MDA/ML-based SEI process that uses the preambles' frequency representation to classify emitters.

Table 7(b) shows FAR and TRR for residual preambles collected at 30 dB and SEI processes trained *with* decoy emitter residual preambles represented within the training set. Using the replay mimicry countermeasure with a HackRF SDR, Eve is rejected at a TRR = 96.15% by the CNN-based SEI process that extracts RF-DNA fingerprints from the emitters' Gabor images. The CNN-based SEI process rejects a HackRF and GAN-based mimicry countermeasure employing Eve at a TRR of 98.06% when extracting RF-DNA fingerprints from the frequency representation and 91.64% when it extracts RF-DNA fingerprints from the Gabor images. When a HackRF-based Eve employs replay, AE, or GAN-based mimicry countermeasures against the MDA/ML-based SEI process that classifies emitters using handcrafted Gabor-based RF-DNA fingerprints, Eve is rejected at TRRs of 94.08%, 93.14%, and 95.64%, respectively. The HackRF-based Eve is rejected at a TRR = 92.85% when employing the GAN-based mimicry countermeasure against the MDA/ML SEI process that classifies emitters using their preambles' frequency representations. When the B210-based Eve uses the replay mimicry countermeasure against the

CNN-based SEI process that extracts RF-DNA fingerprints from the emitters' Gabor images, Eve achieves a FAR=21.48%. The B210-based Eve is rejected at a TRR=90.51% when it employs the GAN-based mimicry countermeasure against the MDA/ML-based SEI process that uses the handcrafted Gabor-based RF-DNA fingerprints.

Table 7(c) shows FAR and TRR for residual preambles collected at 9 dB and SEI processes trained *without* decoy emitter residual preambles represented within the training set. A HackRF-based Eve employing a replay mimicry countermeasure achieves FARs of 14.08%, 18.68%, and 14.44% against the CNN-based SEI processes that extract RF-DNA fingerprints from the time, frequency, and Gabor image representations, respectively. The HackRF-based Eve's AE-based mimicry countermeasure achieves FARs of 16.63% and 20.55% against the CNN-based SEI processes that extract RF-DNA fingerprints from the time and frequency representations, respectively. When the HackRF-based Eve employs the GAN-based mimicry countermeasure, a FAR=12.78% is achieved against the CNN-based SEI process that extracts RF-DNA fingerprints from the residual preambles' frequency representation. The HackRF-based Eve employing replay and AE-based mimicry countermeasures achieves FARs of 14.21% and 14.43% against the MDA/ML SEI process that classifies emitters using the handcrafted Gabor-based RF-DNA fingerprints. For the replay, AE, and GAN-based mimicry countermeasures, the B210-based Eve achieves FARs of 13.46%, 19.46%, and 14.53% versus the CNN SEI process that extracts RF-DNA fingerprints from the residual preambles' time representation. A FAR of 16.39% results when the GAN-based mimicry countermeasure is employed by the B210-based Eve against the frequency representation-reliant CNN-based SEI process. Eve achieves FARs of 14.36% and 18.90% when using either a replay or GAN-based mimicry countermeasure and a B210 against the CNN-based SEI process that extracts RF-DNA fingerprints from Gabor images. FARs of 14.36% and 18.90% are achieved by the B210-based Eve when it employs a replay mimicry countermeasure against MDA/ML-based SEI processes that rely on handcrafted Gabor RF-DNA fingerprint and frequency representations, respectively. The FAR increases to 19.54% when Eve uses a B210 and the GAN-based mimicry countermeasure against the frequency representation-reliant MDA/ML SEI process.

Table 7(d) shows FAR and TRR for residual preambles collected at 9 dB and SEI processes trained *with* decoy emitter residual preambles represented within the training set. The HackRF-based Eve achieves a FAR=13.30% using the AE-based mimicry countermeasure against the CNN-based SEI process that extracts RF-DNA fingerprints from the frequency representations. The HackRF-based Eve is rejected at a TRR of 97.84% and 95.19% when using AE and GAN-based mimicry countermeasures are used against the MDA/ML-based SEI process that performs classification using the residual preambles' frequency representation. When the B210-based Eve employs the replay or GAN-based mimicry countermeasures, the CNN-based SEI process, which extracts RF-DNA fingerprints from the time representation, results in FARs of 14.33% and 13.16%, respectively. The B210-based Eve achieves a FAR=19.63% when employing a GAN-based mimicry countermeasure against the frequency representation-reliant CNN-based SEI process. The B210-based Eve's replay mimicry countermeasure achieves FARs of 14.03% and 20.08% when employing the replay mimicry countermeasure against the MDA/ML SEI processes that classify emitters using handcrafted Gabor-based RF-DNA fingerprints or the frequency representation, respectively. When the B210-based Eve employs the GAN-based mimicry countermeasure, it is rejected at a TRR=92.01% by the MDA/ML SEI process that uses the residual preambles' frequency representations to classify emitters.

When considering the results presented in Tables 6 and 7, Eve's SEI mimicry countermeasures are most effective at 9 dB and when the SEI processes are trained using only the authorized/known emitters' residual preambles (i.e., decoy emitter residual preambles are not used to train any of the SEI processes). This is unsurprising because most SEI processes struggle to separate known emitters as SNR degrades. So, introducing SEI mimicked features and no point of reference (i.e., without knowledge of the decoy emitter's SEI features) only exacerbates the issue. When not using the decoy emitter's preambles or their representations, the SEI process defeats Eve's mimicry countermeasures with the greatest success when it uses the preambles' or residual preambles' frequency representations and the MDA/ML classifier. Lastly, a lower SWaP-C Eve is less effective in hindering or defeating the SEI processes, regardless of the mimicry countermeasure. This may be because the HackRF One SDR is not ideally suited to operate within the IEEE 802.11a Wi-Fi frequency band. Doing so results in additional SEI coloration being imparted upon the transmitted signals.

5.3 Results: "Coffee shop scenario"

In addition to SDR SWaP-C, the "coffee shop scenario" considers two cases; thus, the results for this scenario are presented based on these two cases. Regardless of the case, the success of Eve's SEI mimicry countermeasure is determined by

Table 8 Coffee shop scenario case #1: percent classification for Eve using SEI mimicry and its unmodified signals are used to train the SEI process

(a) Eve using a HackRF One SDR

Target	Actual	User A (%)	User B (%)	User C (%)	Eve (%)
User A	Eve—No GAN	1.88	1.92	1.45	94.75
	Eve—GAN	34.68	1.15	1.44	62.73
User B	Eve—No GAN	0.12	1.92	2.32	95.64
	Eve—GAN	1.84	38.34	1.38	58.44
User C	Eve—No GAN	1.88	1.92	1.45	94.75
	Eve—GAN	0.95	0.12	31.68	67.25

(b) Eve using a B210 SDR

Target	Actual	User A (%)	User B (%)	User C (%)	Eve (%)
User A	Eve—No GAN	0.74	0.38	0.85	98.02
	Eve—GAN	46.50	2.05	4.00	47.45
User B	Eve—No GAN	0.15	0.85	1.35	97.65
	Eve—GAN	1.05	43.80	2.30	52.85
User C	Eve—No GAN	0.74	0.38	0.86	98.02
	Eve—GAN	5.10	2.18	49.64	43.08

“GAN” and “No GAN” indicate Eve’s SEI mimicry countermeasure is “on” and “off”, respectively

calculating the ratio between the number of times the modified preambles—output by Eve’s Location D trained G—are classified as originating from User A and the total number of SEI process decisions. Eve’s success rate is calculated the same way when Eve operates at Location D but mimics the SEI features of User C or Location E and mimics User B.

5.3.1 Coffee shop scenario case #1

The SEI process is trained using Eve’s unmodified signals; thus, it can identify Eve when Eve’s SEI countermeasure is *off*. The AP saves Eve’s preambles transmitted from Location D and Location E using separate files. Case #1 results are presented in Table 8(a) and (b) for a HackRF and B210-based Eve, respectively.

Table 8(b) shows the FAR is 46.5% and 49.64% when Eve launches the SEI mimicry countermeasure—from Location E—using a B210 to mimic User A’s and User C’s SEI features, respectively. From Location D, Eve’s SEI mimicry countermeasure degrades the SEI process’ ability to distinguish Eve from User A and User C by almost 50%. When Eve mimics the SEI features of User B from Location E, the SEI process identifies Eve correctly 43.8% of the time, which is in stark contrast to the 97.65% correct identification rate when Eve is not mimicking User B (a.k.a., “Eve—No GAN”). The SEI process performs better than a guess, but its ability to classify Eve is reduced. Eve’s attack is less successful when operating from Location E, which is attributed to the SEI process’ training set not containing any of Eve’s unmodified signals (a.k.a., SEI mimicry is *off*) transmitted from Location E. We suspect the success rate would change if this were not the case.

When Eve uses a HackRF SDR (see Table 8(a)), the SEI process accurately classifies Eve’s signals at a percent correct classification rate of 94.75%, 94.75%, and 95.64% when Eve is *not* mimicking (a.k.a., “Eve—No GAN”) the SEI features of User A, User B, and User C, respectively. Thus, the SEI process easily distinguishes Eve from the authorized users. However, once Eve employs an SEI mimicry countermeasure (a.k.a., “Eve—GAN”), the SEI process’s accuracy is degraded. When mimicking User A’s SEI features, the SEI process’s ability to distinguish Eve from User A is reduced by 34.68%; thus, Eve is correctly identified 62.73% of the time. When mimicking User C, Eve’s GAN-based mimicry countermeasure reduces the SEI process accuracy from 94.75 to 67.25%; thus, Eve’s signals are being misclassified as originating from User C 31.68% of the time. Eve’s mimicry of User C is the least effective, while Eve’s SEI mimicry countermeasure is most effective when Eve mimics User B’s SEI features. When Eve mimics User B, SEI process accuracy is reduced from 95.64 to 58.44%. Overall, Eve’s SEI countermeasure hinders the SEI process’s ability to distinguish Eve from the targeted emitter, but not to the extent that they are indistinguishable. We also note that the results in Table 8(a) are poorer than those in Table 8(b) that correspond to a USRP B210-based Eve. The reduced performance—exhibited in Table 8(a)—is attributed to the SWaP-C

Table 9 Coffee shop scenario case #2: percent correct classification for Eve using SEI mimicry and its unmodified signals are not used to train the SEI process

(a) Eve using a HackRF one SDR

Target	Actual	User A (%)	User B (%)	User C (%)
User A	Eve—No	32.25	35.25	32.16
	Eve—GAN	74.18	10.37	15.45
User B	Eve—No	33.46	31.26	35.28
	Eve—GAN	6.34	77.06	16.58
User C	Eve—No	32.59	35.25	32.16
	Eve—GAN	18.28	10.34	71.38

(b) Eve using a B210 SDR

Target	Actual	User A (%)	User B (%)	User C (%)
User A	Eve—No GAN	37.62	32.32	30.06
	Eve—GAN	80.65	14.05	5.30
User B	Eve—No GAN	25.79	32.96	41.25
	Eve—GAN	11.35	67.04	21.25
User C	Eve—No GAN	37.62	32.32	30.06
	Eve—GAN	7.35	3.20	89.45

“GAN” and “No GAN” indicate Eve’s SEI mimicry countermeasure is “on” and “off”, respectively

constraints imposed on Eve when using a HackRF instead of a B210. The most prominent constraint is a lower sampling rate of 20 MHz versus the B210’s 40 MHz. The sampling rate had to be lowered to keep the HackRF from ceasing transmissions once it connected to the NVIDIA Jetson Nano. If the sampling rate is not reduced, most of the collected Wi-Fi frames’ samples are missing, negatively impacting preamble detection and extraction.

5.3.2 Coffee shop scenario case #2

The SEI process *is not* trained using Eve’s unmodified signals. The SEI process CNN is trained using only authorized emitter signals, and Eve is unknown before employing its mimicry countermeasure. Table 9 shows Eve is classified as each user about one-third of the time when its mimicry countermeasure is *off*. The CNN is guessing which authorized user transmitted the signals; thus, there is no bias toward one user over another that would bias the results when Eve mimics each user’s SEI features.

When Eve mimics User A’s SEI features, the SEI process classifies them as originating from User A at a 74.18% accuracy, Table 9(a). The remaining signals are classified as originating from User B and User C at accuracies of 10.37 and 15.45%, respectively. When mimicking User B, Eve’s SEI countermeasure results in its modified signals being classified as originating from User B at an accuracy of 77.08%; this represents Eve’s highest accuracy when using a HackRF to implement the SEI mimicry countermeasure and is 9.68% higher than Eve’s accuracy when using a B210, Table 9(b). The HackRF-based Eve’s lowest accuracy of 71.38% occurs when mimicking User C, Table 9(a). However, the B210-based Eve results in a lower classification accuracy of 67.4% when mimicking User B’s SEI features, Table 9(b). This performance difference may be due to the HackRF One’s lower sampling rate. A lower sampling rate means some SEI nuances are missing from the emitters’ sampled signals, which impedes the SEI process’ ability to discriminate between emitters. This is especially important when the SEI process is trained without Eve’s unmodified signals because the CNN needs as much information as possible to achieve reliable discrimination.

Table 9(b) shows the SEI countermeasure success rate for the B210-based Eve. When Eve transmits with the SEI mimicry countermeasure *turned off* (a.k.a., “Eve—No GAN”), the SEI process is guessing which emitter transmitted the signals. However, once Eve uses the SEI mimicry countermeasure (a.k.a., “Eve—GAN”), the SEI process mostly assigns Eve the identity of the mimicked user. The SEI mimicry countermeasure’s greatest success occurs when User C is mimicked, which results in an 89.45% identification rate versus only 30.1% with the countermeasure *turned off*. The SEI countermeasure’s lowest success rate is 67.4% and occurs when Eve mimics the SEI features of User B. This is roughly a 44% increase in the identification of Eve as User B. This is interesting when considering that when Eve operates at

Table 10 FAR and TRR—presented in each cell as “FAR/TRR” using percentages—when a DAE precedes the SEI process’ trained using a decoy emitter’s RF-DNA fingerprints at an SNR 9 dB

		CNN			MDA/ML	
		Time	Frequency	Gabor	Gabor	Frequency
HackRF	RE	17.59%/22.59%	12.50%/00.00%	12.56%/00.00%	08.61%/08.60%	12.49%/00.35%
	AE	11.34%/40.79%	12.50%/00.00%	12.55%/00.00%	12.14%/07.20%	12.50%/00.16%
	GAN	08.54%/34.86%	12.50%/00.00%	12.19%/00.00%	11.74%/06.73%	12.48%/00.23%
B210	RE	02.04%/69.66%	12.50%/00.00%	12.61%/00.00%	17.65%/03.34%	00.09%/99.73%
	AE	00.95%/73.81%	12.50%/00.00%	17.18%/00.01%	07.15%/23.38%	00.03%/98.06%
	GAN	00.98%/73.09%	12.50%/00.00%	14.85%/00.00%	12.78%/30.00%	00.00%/99.79%

SEI processes that result in a FAR above 12.5% (a.k.a., a guess) and a TRR above 90% are marked using italicized and bold font, respectively. “RE”, “AE”, and “GAN” indicate Eve’s replay, AE, and GAN-based mimicry countermeasures

Location E with its SEI mimicry countermeasure *turned off*, the SEI process identifies Eve as User B roughly 33% of the time. This is not the highest rate—the highest rate is 41.25%, corresponding to User C—which shows that the SEI mimicry countermeasure alters Eve’s SEI features sufficiently to change which user Eve “looks most like”. Eve goes from “looking most like” User C to “looking most like” User B when the SEI mimicry countermeasure goes from *off* to *on*.

5.4 Results: Denoised signals with MDA/ML-based SEI

Motivated by our work in [14], a DAE is added to the SEI process to improve rejection of Eve’s SEI mimicry countermeasures at an SNR or 9 dB. The DAE is trained using 30 dB and 9 dB collected preambles with the 30 dB preambles serving as the targeted output. The 9 dB preambles are matched to the same emitter’s 30 dB preambles. At an SNR=9 dB, Eve’s mimicry countermeasures are detected at the highest rate when the SEI process learns the decoy emitter’s RF-DNA fingerprints, Table 10. When employing the replay-based mimicry countermeasure, a HackRF-based Eve achieves FARs of 17.59% and 12.56% against a CNN-based SEI process that uses the time representation and handcrafted Gabor-based RF-DNA fingerprints, respectively. A FAR=12.55% results when a HackRF-based Eve employs an AE-based mimicry countermeasure against the CNN-based SEI process that extracts RF-DNA fingerprints from the Gabor images. However, this same CNN-based SEI process bor-based corresponds to FARs of 12.61%, 17.18% and 14.85% when Eve uses a B210 with replay, AE, and GAN-based mimicry countermeasures, respectively. FARs of 17.65% and 12.78% result when Eve uses a B210 with a replay or a GAN-based mimicry countermeasure against the MDA/ML-based SEI process that uses handcrafted Gabor-based RF-DNA fingerprints. However, the MDA/ML-based SEI process achieves TRRs of 99.73%, 98.06%, and 99.79% against a B210-based Eve that uses a replay, AE, or GAN-based mimicry countermeasure, respectively.

6 Conclusion

This work considers a strong adversary (a.k.a. Eve) that uses “off-the-shelf” software tools, DL, and SDR to perform digital and SEI mimicry to deceive an SEI-based monitor (a.k.a. Bob) to gain network access. SEI processes employing classification without knowledge of Eve’s organic features or those of a representative (a.k.a., a decoy) incorrectly authenticate Eve as an authorized user. The success of Eve’s SEI mimicry countermeasure is higher when Eve uses a higher SWaP-C SDR (e.g., a B210 versus HackRF One). FAR is higher at 9 dB than at 30 dB; thus, it is advantageous for Eve to launch SEI mimicry at lower SNRs to increase the effectiveness of its countermeasure and confusion of the SEI process. Eve can achieve a lower SNR by intentionally adding noise to its signals before transmission or decreasing its transmit power. The effectiveness of Eve’s SEI mimicry countermeasure is degraded when the SEI process is trained using decoy emitter signals. TRR is lower for all SEI processes, both SDRs and the three SEI mimicry countermeasures at 9 dB. Denoising signals before SEI does not decrease the success of Eve’s SEI mimicry countermeasures. This may be due to the DAE “washing out” essential emitter-specific features used by the SEI process to distinguish between emitters. Overall, handcrafted SEI—without DAE—using Gabor-based RF-DNA fingerprints provides the best FAR/TRR performance, albeit at a more significant computational cost. Many of these observations are validated by assessing Eve’s SEI mimicry countermeasure in a “coffee shop” setting. Future work will consider open-set SEI to remove the need for decoy emitter signals and one-to-one identity verification to overcome classification’s “forced decision” drawback.

7 Future work and other considerations

The presented work provides an initial investigation into the feasibility, ease at which, and technological knowledge, hardware, and software “entry point(s)” needed by an adversary to implement SEI mimicry countermeasures. However, it is not comprehensive, and future research is required. For instance, the number of emitters and size of the “coffee shop” are limited. So, future investigations into larger operational environments that include mobility and networks comprised of a more significant number of emitters are worth pursuing to determine the effectiveness of SEI mimicry in such cases and to develop SEI processes that are robust to SEI mimicry. Initial research indicates that increasing the number of emitters will prove advantageous to the SEI mimicking adversary because misclassification rises as the number of emitters increases [2].

One possible method for countering SEI mimicry is open-set SEI. It may remove the need for decoy emitter signals by providing the SEI process with an “other” or “catch-all” case to assign unknown or previously unseen signals. Another possible method that may increase the SEI process’ robustness to SEI mimicry countermeasures is one-to-one identity verification because it overcomes classification’s “forced decision” drawback [11, 42]. Investigating adversarial learning may also prove beneficial to increasing an SEI process’s robustness to SEI mimicry.

The previously proposed methods focus on detecting the attacker using SEI, but SEI intends to augment other existing and proposed security solutions. So, investigating combining SEI and other security approaches is a logical next step. As stated in Sect. 4.3 the channel between the SEI mimicking adversary (a.k.a., Eve) and the SEI-based authenticator (a.k.a., Bob) is unique versus the channels that exist between Bob and the authorized network users (i.e., User A, User B, and User C in our case). The combination of SEI with Channel State Information (CSI) would investigate using the unique emitter features and unique channel characteristics to overcome SEI mimicry countermeasures while simultaneously overcoming CSI’s “first transmission assumption,” which assumes two communicating entities are sufficiently known to one another using another identification scheme (e.g., encryption [43]). CSI is just one example and there may be others that prove advantageous and beneficial to defeating SEI mimicry.

Author contributions J.T. set up and conducted all signal collection and preprocessing not associated with the coffee shop scenario. J.T. also generated and analyzed all deep learning-related SEI results and configuration and implementation of the adversary and its associated countermeasures. M.F. performed all experiments, data collection, and analysis associated with the coffee shop scenario portion of the paper. M.F. also guided the deep learning architecture configurations and training. M.H. conducted all of the SEI work associated with the handcrafted RF-DNA fingerprinting and the MDA/ML classifier. D.R. originated the idea of the research, provided oversight and guidance of the research, and edited the manuscript. D.L. guided the research and reviewed and edited the paper. J.T., M.F., and D.R. wrote the initial draft of the paper.

Data availability The data supporting this study’s findings are not openly available due to sensitivity and third-party restrictions.

Declarations

Competing interests The authors declare no competing interests.

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